



The power sector is banking on digitalization, which will bring in accuracy and precision, and make the entire power generation process more accountable while checking wastage. Big data, virtual power plants, demand response technologies, smart grids, IoT, AI, blockchain, use of drones can create a more flexible and efficient electricity system that delivers energy seamlessly across various consumer segments in the future.

Tata Power is enabling customers to be a part of its digital drive. As part of its smart grid technologies which promote energy savings, the company has initiated smart metering for customers in Mumbai and Delhi, where over 65,000 smart meters have already been installed. As a joint initiative between Tata Power, Tata Trusts, and the Government of India, the country's first-of-its-kind clean energy lab has also been established for startups, entrepreneurs and innovators, offering complete 'lab to market' incubation support to clean energy enterprises. The Company and Social Alpha have joined hands to build a unique innovation curation and venture development program 'Net-Zero Industry Accelerator' focused on industrial use cases of clean energy transition and industrial decarbonization contributing towards India's net-zero targets.

The Company is committed to leveraging technology in new-age businesses like EV Charging, Solar Rooftop, Home automation, and creating innovative low-carbon solutions for customers.

rection, with lesser dependency on non-renewable sources, with the eventual goal of achieving decarbonization.

Future Ready Transition

With sustainability as the goal, companies must adopt the right technologies and digital tools that will ensure that they achieve it in time.



All-round mission

One of the fundamental aims of sustainability is to create a circular economy, by preventing wastage of resources and promoting recycling. Efficient waste management must be a norm across the value chain, to make the entire process sustainable, thus resulting in circularity. Setting up the blocks to contribute to a circular economy, Tata Power has taken significant strides in proactive waste management practices across its value chain, including maximization of fly-ash utilization and progressing to zero waste to landfill (in cases of PV panel disposal). Besides, it is adopting practices that will make it 100% water neutral before 2030.

Meanwhile, Tata Power strongly believes in resource conservation, energy efficiency, habitat protection, and the enrichment and development of local communities in and around areas of operations. For instance, Tata Power's hydro sites in Maharashtra have been in continuous harmony with nature for over 100 years in the Western Ghats, with the region surrounding the sites housing rich biodiversity. Also, with support from its employees, Tata Power is running a volunteering program (Arpan) which oversees multiple biodiversity conservation efforts, including Tree Mittra (to conserve native species) and Club Enerji (to develop young conservation champions). The Company is also undertaking a GIS mapping in our hydropower plant areas which will help in assessing efforts toward carbon mitigation.

As part of its long-term efforts, the Mahseer conservation program, which was taken up in 1970 for ecological enrichment of the hydel lakes and to rehabilitate the Deccan Mahseer in Maharashtra, has helped the fish be declared as 'least concern' species in the IUCN red list. Besides, in Odisha, awareness is being created around elephant conservation, with 427 DTR fencing being constructed for public safety and the elephant corridor.

The larger premise of sustainability is to improve people's lives in tandem with the conservation of nature. Tata Power is constantly striving to strengthen ecosystems for inclusive education, digital and financial literacy, skilling, and livelihood of people within the community, through a variety of schemes such as Anokha Dhaaga, Abha, and Adhikaar, among others.

Through a range of varied initiatives to foster sustainability, and with industry doing its bit, India is poised for a green future, but the 'power' to change the world rests with us, the people.